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ITM SLS BARODA UNIVERSITY, VADODARA

Course: JavaScript Framework (PBL) - C2620A3
Program: BTech Semester 6 | Academic Year 2026–27

A PROJECT REPORT ON

SMART COLLEGE ATTENDANCE SYSTEM WITH FACE RECOGNITION

Automated Biometric Attendance Management, Anti-Proxy Verification & Defaulter Analytics

Project Title:	Smart College Attendance System with Face Recognition
Group Code:	GRP-C2620A3-04 (Class 4)
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Semester & Section:	5th sem class 4 (Academic Year 2026–27)
Guided By:	Faculty Guide & Project Coordinator Department of B.TECH CSE(AI)
Institution:	School of Computer Science & Engineering ITM SLS Baroda University, Vadodara, Gujarat

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CERTIFICATE

This is to certify that the project report entitled **“Smart College Attendance System with Face Recognition”** submitted by the students of BTech Semester 6 for the course **“JavaScript Framework (PBL) - C2620A3”** during Academic Year 2026–27 is a record of project work carried out under the guidance of the undersigned and submitted in partial fulfilment of the academic requirements of the course.

Guide: _____ **Date:** _____

Department/School: Department of B.TECH CSE(AI) **Signature:** _____

Head of Department: _____ **Seal:** _____

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DECLARATION

We hereby declare that the project entitled **“Smart College Attendance System with Face Recognition”** is our original academic work prepared for the course **“JavaScript Framework (PBL) - C2620A3”**. All external sources, libraries, frameworks, documentation, and other materials used in the project have been appropriately acknowledged. The report and implementation have not been copied or submitted elsewhere for academic credit.

Date: October 2026
Place: Vadodara, Gujarat

1. Vishakha (Enrollment No: 24C22075)
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Signature: _____

ACKNOWLEDGEMENT

We express our deepest gratitude to our respected Faculty Guide, Course Coordinator for **JavaScript Framework (PBL) - C2620A3**, and the Head of the Department of B.TECH CSE(AI), ITM SLS Baroda University, Vadodara, for their valuable mentorship, constructive feedback, and encouragement throughout this project. We also thank the university administration for providing the necessary computing resources and lab infrastructure to implement and test this practical system.

ABSTRACT

Traditional classroom attendance logging through paper rosters takes 10 to 15 minutes of lecture time and is severely compromised by proxy attendance. This project presents a practical, automated **Smart College Attendance System** developed as a project-based learning (PBL) initiative for the course *JavaScript Framework (PBL) - C2620A3*. The system is built as a complete full-stack JavaScript application using **React 18** on the frontend, **Node.js & Express.js** on the backend, **MongoDB** as the NoSQL database, and **face-api.js** for real-time 128-dimensional biometric facial embedding extraction in the browser. It supports rapid 1 or 2 photo biometric enrollment, eliminates duplicate attendance logging within the same lecture, strictly rejects unauthorized individuals with a red alert HUD, and automates semester examination eligibility compliance (< 75% attendance threshold) with one-click Excel and CSV reporting.

1. INTRODUCTION & PROBLEM STATEMENT

In university classrooms, regular attendance monitoring is directly linked to academic performance and accreditation compliance (NAAC and NBA). University regulations require students to maintain at least **75% attendance** to qualify for semester end examinations. However, traditional manual methods suffer from significant limitations:

- **Time Overhead:** Calling 60+ roll numbers wastes 10 to 15 minutes of every lecture hour.
- **Proxy Attendance:** Friends mark presence for absentees, making attendance records inaccurate.
- **Clerical Workload:** Teachers manually calculate percentages at the end of every month.
- **Late Defaulter Warnings:** Students only discover they have < 75% attendance just days before exams.

Project Objective: Build a responsive biometric attendance system running on standard classroom computers that verifies student identity in under 1 second, prevents duplicate marking, and produces instant analytics without requiring cloud fees.

2. SYSTEM ARCHITECTURE & DATA FLOW

The application is engineered as a modern full-stack JavaScript architecture consisting of three layers:

1. **Frontend Layer (React 18 & Vite):** A responsive Single Page Application (SPA) built with React, Tailwind CSS, Lucide Icons, and HTML5 <video> / <canvas> rendering real-time biometric HUD bounding boxes directly in the browser.
2. **Biometric Face Recognition Engine:** Built with @vladmandic/face-api (TensorFlow.js) computing 128-dimensional facial embedding vectors locally with sub-50ms latency per frame.
3. **Backend API & Persistence Layer:** A Node.js & Express REST API communicating with a local MongoDB database via Mongoose ODM, utilizing compound unique indexes to guarantee anti-proxy duplicate prevention.

3. METHODOLOGY & BIOMETRIC ENGINE

The face recognition pipeline is engineered around deep neural network embeddings executing 100% in JavaScript:

- **SSD MobileNet V1 & Tiny Face Detector:** Neural network weights loaded from local models directory localize faces in video streams at 25+ frames per second.
- **68-Point Facial Landmark Mapping:** Pinpoints facial geometry across nose bridge, eye pupils, lips, and jaw contours for rotation-invariant tracking.
- **128-Dimensional Deep Descriptor Extraction:** Converts biometric facial features into a 128-element normalized vector stored directly in MongoDB.
- **Euclidean Distance Matching:** Compares live video descriptors against MongoDB student vectors using Euclidean distance. Distance < 0.55 triggers a genuine match (≥ 65% confidence) with green HUD box; distance ≥ 0.55 triggers red 'Unknown Face' rejection.

4. MONGODB DATABASE DESIGN & SCHEMA

The database is managed locally in MongoDB (mongodb://127.0.0.1:27017/college_attendance) and comprises two Mongoose schemas:

Collection: students	Collection: attendances
• **_id**: ObjectId (Primary Key) • **rollNo**: String (Unique, e.g. 24C22075) • **name**: Full Student Name • **department**: B.TECH CSE(AI) • **semester**: Number (5 / 6) • **section**: Class 4 • **faceDescriptors**: [[Number]] (128-d arrays) • **photoSample**: Base64 thumbnail	• **_id**: ObjectId (Primary Key) • **studentId**: ObjectId ref -> students • **rollNo**: Student Roll Number • **name**: Student Name • **date**: Lecture Date (YYYY-MM-DD) • **time**: Scan Timestamp (HH:MM:SS) • **course**: Subject Name • **confidence**: Biometric Match % • *Index*: UNIQUE(rollNo, date, course)

5. TESTING, EXPERIMENTAL RESULTS & EVIDENCE

The project was verified using automated Node.js test suites testing MongoDB connections, student registration, 128-d descriptor persistence, duplicate suppression, and 75% defaulter aggregation. **All 5 tests passed with 100% success rate.**

Live face scanning experiments conducted with registered student faces and unknown individuals produced the following results:

Test Scenario	Match %	System Action	Outcome
Registered Student (Vishakha - Direct Front)	100.0%	Green HUD • Marked Present	Pass [Correct Match]
Registered Student (Vishakha - Angular View)	76.5%	Green HUD • Marked Present	Pass [Correct Match]
Registered Student (Jithendar - Front View)	92.4%	Green HUD • Marked Present	Pass [Correct Match]
Duplicate Scan in Same Class Period	76.5%	Blocked (409 Already Marked)	Pass [Anti-Duplicate]
Unregistered Individual / Stranger	54.0%	Red HUD • 'Unknown Face'	Pass [Rejected Proxy]

6. TEAM CONTRIBUTION & INDIVIDUAL ROLES

Team Member	Enrollment No	Key Responsibilities & Modules Delivered
Vishakha	24C22075	• React 18 SPA Architecture & UI Component Design (Tailwind, Lucide, Canvas) • Face Recognition Engine integration via @vladmandic/face-api (128-d vectors) • Automated testing suite and biometric client-side matching verification
Jithendar	24C22074	• Node.js & Express.js REST API server endpoints (Port 5000) • MongoDB Mongoose schema design with UNIQUE(rollNo, date, course) constraint • Attendance aggregation logic for university 75% examination compliance and CSV export

7. DEPLOYMENT & SYSTEM ADVANTAGES

1. **Saves Teaching Time:** Eliminates 10 to 15 minutes of wasted time per lecture, giving faculty more time to instruct.
2. **Zero Hardware Expense:** Runs directly on standard laptops or desktop PCs with built-in or USB webcams. No costly biometric terminals required.
3. **Zero Cloud Fees:** 100% offline edge processing without monthly subscription fees or third-party cloud data risks.
4. **Audit & Accreditation Compliance:** Generates instant timestamped attendance logs ready for university examinations and NAAC/NBA inspections.

8. CONCLUSION

The **Smart College Attendance System with Face Recognition** successfully demonstrates how computer vision and modern web frameworks can be integrated to resolve practical campus challenges. Developed as part of **JavaScript Framework (PBL) - C2620A3** at **ITM SLS Baroda University, Vadodara**, the system achieves sub-50ms recognition latency, blocks proxy attendance, and automates defaulter tracking cleanly and effectively.

9. REFERENCES

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